```
import java.util.Scanner;

public class WarehouseInventoryScanner {

    public static void main(String[] args) {
        int[][] warehouse = {
            {15, 30, -1, 8},
            {4, -1, 25, 12},
            {50, 2, 18, -1}
        };

        Scanner scanner = new Scanner(System.in);
        int choice;

        do {
            System.out.println("\n--- Warehouse Inventory Menu ---");
            System.out.println("1. Calculate Total Inventory");
            System.out.println("2. Find First Critically Low Shelf");
            System.out.println("3. Simulate Restocking");
            System.out.println("4. Exit");
            System.out.print("Enter choice: ");
            choice = scanner.nextInt();

            if (choice == 1) {
                int total = 0;
                for (int i = 0; i < warehouse.length; i++) {
                    for (int j = 0; j < warehouse[i].length; j++) {
                        if (warehouse[i][j] == -1) {
                            continue;
                        }
                        total += warehouse[i][j];
                    }
                }
                System.out.println("Total inventory count: " + total);

            } else if (choice == 2) {
                int lowThreshold = 5;
                int foundRow = -1;
                int foundCol = -1;
                boolean found = false;

                for (int i = 0; i < warehouse.length; i++) {
                    for (int j = 0; j < warehouse[i].length; j++) {
                        if (warehouse[i][j] >= 0 && warehouse[i][j] <= lowThreshold) {
```

```
                    foundRow = i;
                    foundCol = j;
                    found = true;
                    break;
                }
            }
            if (found) {
                break;
            }
        }

        if (found) {
            System.out.println("First critically low shelf found at Row " + foundRow + ", Column
" + foundCol + " (Items: " + warehouse[foundRow][foundCol] + ")");
        } else {
            System.out.println("No critically low shelves found.");
        }

    } else if (choice == 3) {
        simulateRestock(warehouse, scanner);
    }

} while (choice != 4);

scanner.close();
}

public static void simulateRestock(int[][] warehouse, Scanner scanner) {
    while (true) {
        System.out.print("Enter row index (0-" + (warehouse.length - 1) + "): ");
        int r = scanner.nextInt();
        System.out.print("Enter column index (0-" + (warehouse[0].length - 1) + "): ");
        int c = scanner.nextInt();

        if (r >= 0 && r < warehouse.length && c >= 0 && c < warehouse[0].length) {
            if (warehouse[r][c] == -1) {
                System.out.println("Error: Cannot restock a damaged shelf.");
                return;
            }

            System.out.print("Enter restock quantity: ");
            int quantity = scanner.nextInt();

            if (quantity > 0) {
```

```
                warehouse[r][c] += quantity;
                System.out.println("Restocked successfully. New count: " + warehouse[r][c]);
                return;
            }
        }
        System.out.println("Invalid layout position or quantity. Please re-enter.");
    }
  }
}
```

OUTPUT GENERATED FILES DEBUGGER GUI

```
--- Warehouse Inventory Menu ---
1. Calculate Total Inventory
2. Find First Critically Low Shelf
3. Simulate Restocking
4. Exit
Enter choice: 1
Total inventory count: 164

--- Warehouse Inventory Menu ---
1. Calculate Total Inventory
2. Find First Critically Low Shelf
3. Simulate Restocking
4. Exit
Enter choice: 2
First critically low shelf found at Row 1, Column 0 (Items: 4)
```

OUTPUT GENERATED FILES DEBUGGER GUI

```
First critically low shelf found at Row 1, Column 0 (Items: 4)

--- Warehouse Inventory Menu ---
1. Calculate Total Inventory
2. Find First Critically Low Shelf
3. Simulate Restocking
4. Exit
Enter choice: 3
Enter row index (0-2): 1
Enter column index (0-3): 2
Enter restock quantity: 23
Restocked successfully. New count: 48

--- Warehouse Inventory Menu ---
1. Calculate Total Inventory
2. Find First Critically Low Shelf
3. Simulate Restocking
4. Exit
Enter choice: 4
```